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RESEARCH ARTICLE

CAJU: Convolutional Attentive Joint Units for Infectious Keratitis In Vivo Confocal Microscopy Image Classification

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ABSTRACT Infectious keratitis (IK) is a significant cause of preventable vision loss and necessitates expeditious, dependable etiological assessment. Conventional diagnostics (e.g., corneal scraping, culture, and molecular tests) are often invasive, time-consuming, and resource-dependent, which limits their practicality in routine workflows. As a non-invasive alternative, in vivo confocal microscopy (IVCM) enables high-resolution visualization of corneal microstructures and potential pathogens. However, its interpretation requires specialized expertise and remains subject to inter-observer variability. We hereby propose Convolutional Attentive Joint Units (CAJU), an attention-guided dual-stream pipeline for IK classification from IVCM images. CAJU instantiates complementary CNN backbones and fuses their representations through a transformer-inspired self-attention module, allowing backbone selection according to performance and deployment constraints. The public IVCM-Keratitis dataset was subjected to a series of experiments following the implementation of artifact filtration, encompassing a total of 2,155 images classified into four distinct categories: AK, FK, NSK, and Normal. These experiments were executed through the utilization of a cross-validation approach. Among the evaluated instantiations, CAJU (DenseNet-161 + ResNet-101, w/ SA) demonstrated optimal class-balanced performance, attaining 96.22% precision, 96.01% recall, and 96.03% F1-score, while CAJU (VGG16 + MobileNetV3, w/ SA) attained the maximum accuracy of 97.13%. In order to support interpretability, the present study integrates Grad-CAM and SHAP to localize salient regions and provide class-wise attributions, thereby enabling plausibility checks in both correct predictions and clinically relevant error modes (notably NSK ambiguity). In summary, CAJU offers a dependable and comprehensible framework to facilitate IVCM-based IK diagnosis and to support clinical decision-making, particularly in non-specialized settings.

INDEX TERMS Deep learning, infectious keratitis, attention, classification.

I. INTRODUCTION

Infectious keratitis (IK) is a severe corneal infection and a leading cause of blindness worldwide, especially in developing regions [1]. It manifests as corneal inflammation triggered by pathogenic agents, such as fungi (FK), bacteria (BK), viruses (VK), and *Acanthamoeba protozoa* (AK) [2]. Major risk factors include contact lens contamination, ocular trauma, and immunosuppressive therapies. Without prompt diagnosis and treatment, IK can lead to irreversible vision loss [3].

Standard diagnostic approaches, including corneal scraping followed by microbial culture, are widely used for IK detection due to their specificity. However, these methods are invasive, time-consuming, and may take up to 72 hours to deliver results [4]. As a faster and non-invasive alternative, *in vivo* confocal microscopy (IVCM) enables real-time, high-resolution imaging of the corneal layers, supporting

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the morphological assessment of pathogenic structures [4]. Despite its clinical potential, IVCM image interpretation remains heavily reliant on specialized expertise and subjective visual analysis.

The field of artificial intelligence (AI), particularly the subset known as deep learning (DL), has emerged as a promising solution for automating image-based medical diagnosis. Convolutional neural networks (CNNs), in particular, have demonstrated excellent performance across various imaging modalities (X-ray, MRI, retinal imaging, and more) due to their ability to extract hierarchical patterns from visual data [4], [5]. When applied to ophthalmology, CNNs have the potential to expedite and standardize diagnostic procedures.

In recent years, several studies have explored these capabilities specifically for IVCM analysis. A representative approach integrates data augmentation with histogram-matching image fusion to enhance diagnostically relevant structures, such as hyphae. This approach facilitates high accuracy with lightweight CNN architectures [6]. In a similar vein, a deep learning algorithm based on ResNet-101 was proposed for the classification of fungal keratitis (FK). This algorithm has demonstrated utility not only in the diagnosis of the condition but also in the monitoring of residual fungal structures during treatment [7].

To address the non-explainable nature of deep learning, [8] highlighted the importance of Explainable AI (XAI). Their study showed that integrating visual explanations via Grad-CAM significantly improved the diagnostic accuracy of less experienced ophthalmologists. Specific to *Acanthamoeba* keratitis (AK), [9] developed a comprehensive decision-support system that filters artifacts and classifies corneal layers before identifying infections, validating the potential of multi-stage AI pipelines.

Regarding hybrid architectures, which are central to our proposal, [10] introduced a structure-aware CNN (SACNN) that fuses global features with structure-enhanced inputs, effectively improving feature discrimination. Most relevant to the current study, [3] established the IVCM-Keratitis dataset and benchmarked ten different CNN architectures, identifying DenseNet161 as a strong baseline for this task.

However, despite these advances, the application of CNNs to IVCM imaging faces persistent challenges. These include the limited availability of annotated datasets, class imbalance, image artifacts, and overfitting in small sample scenarios [3]. Additionally, conventional CNNs may not fully capture the subtle and diverse morphological features of IK across different microbial classes. Recent research highlights the potential of hybrid models and attention mechanisms to enhance feature representation and interpretability [10].

To address these challenges, this study proposes Convolutional Attentive Joint Units (CAJU), a flexible dual-stream pipeline for infectious keratitis (IK) classification. The CAJU architecture fuses complementary representations extracted from two CNN backbones and integrates them through a transformer-inspired self-attention (SA) block. The SA module recalibrates the joint embedding by emphasizing diagnostically informative cues while suppressing spurious or low-signal patterns, thereby improving feature fusion beyond direct concatenation.

All experiments are conducted on the public IVCM-Keratitis dataset, which comprises 3,018 labeled grayscale images across four classes: Acanthamoeba keratitis (AK), fungal keratitis (FK), non-specified keratitis (NSK), and healthy corneas. To ensure consistent data quality, non-interpretable frames dominated by saturation or near-darkness are automatically excluded prior to training. In addition to strong quantitative performance, CAJU provides interpretability through complementary explanation methods: Grad-CAM for spatial localization of salient regions and SHAP for finer-grained, input-level attribution, supporting the qualitative assessment of challenging differentials such as FK versus NSK in a decision-support setting.

The main contributions of this study can be summarized as follows:

- A flexible dual-stream pipeline (CAJU) is introduced to learn complementary representations by pairing a texture-oriented backbone with a structure-oriented backbone. CAJU is instantiated and comparatively evaluated on three backbone pairs, as opposed to relying on a single fixed pairing. The first configuration is DenseNet-161 + EfficientNet-B0, the second is DenseNet-161 + ResNet-101, and the third is VGG16 + MobileNet-V3. The objective of this design is twofold: first, to capture both fine-grained pathogen-related cues and broader corneal morphology; and second, to mitigate the limitations of single-backbone models.
- A self-attention (SA) block is employed to adaptively recalibrate the fused feature representation. A dedicated ablation study (with vs. without SA) quantifies its contribution over direct concatenation, showing that SA improves feature fusion and enhances discriminative performance under challenging, noise-prone IVCM conditions.
- Gradient-weighted Class Activation Mapping (Grad-CAM) is employed to provide visual explanations of the model decisions [11]. These saliency maps facilitate clinical interpretability by accentuating diagnostically pertinent regions (e.g., structures consistent with hyphae or cyst-like patterns) and enhancing transparency in assisted diagnoses.
- In addition to Grad-CAM, SHAP-based explanations are incorporated to provide more fine-grained, input-level attributions, enabling qualitative inspection of competing evidence across classes and supporting the analysis of borderline cases such as FK versus NSK.
- All experiments are conducted on the open-source IVCM-Keratitis dataset [3], a publicly available collection of in vivo confocal microscopy images for the infectious keratitis classification.

Importantly, this study addresses recurrent gaps in prior deep learning approaches for IVCM-based keratitis analysis by jointly emphasizing interpretability, computational characterization, and architectural exploration.

Unlike generic dual-stream models, CAJU introduces a modular pipeline designed to exploit morphological complementarity. By pairing texture-oriented backbones with structure-oriented ones, the architecture specifically targets the multiscale nature of infectious keratitis cues in IVCM images. Within this modular framework, the self-attention mechanism acts as a dynamic recalibrator, allowing the model to prioritize the most informative features from each stream on an instance-by-instance basis.

Beyond reporting classification performance, CAJU integrates complementary explanation strategies: Grad-CAM for spatial localization [11] and SHAP for finer-grained input-level attribution to improve transparency and support qualitative inspection of challenging cases. Moreover, rather than committing to a single design, we systematically evaluate multiple complementary backbone pairings within the same dual-stream and self-attention fusion framework, covering both efficient and higher-capacity configurations (VGG16+MobileNetV3, DenseNet-161+EfficientNet-B0, and DenseNet-161+ResNet-101). This design enables an informed trade-off analysis across predictive performance (accuracy, precision, sensitivity, and F1-score) and practical deployment constraints (parameters, model size, GFLOPs, and inference latency). Finally, training and evaluating CAJU on a public dataset enhances reproducibility and provides a consistent reference point for future extensions and external validation.

II. PROPOSED METHOD

As illustrated in Figure 1, the workflow of our study includes the removal of artifact images, image splitting, dataset augmentation, cross-validation, and the evaluation process.

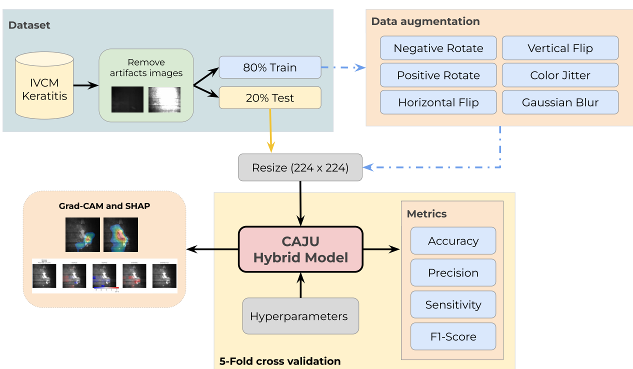

FIGURE 1. The following presentation offers a detailed exposition of the CAJU model's workflow. The IVCM-Keratitis dataset undergoes a series of preprocessing steps that include the removal of artifacts, the division into training and test sets, and the resizing of images. The augmentation of data is implemented on training images. The model is trained using 5-fold cross-validation, and performance is evaluated using classification metrics.

A. DATASET

For this study, the IVCM-Keratitis dataset [12] was selected due to its public availability [3]. The original version of the dataset used in this work comprises 3,018 JPEG grayscale images with a resolution of 768×576 pixels, categorized into four classes: 1,391 images of AK, 863 of FK, 231 of non-specified keratitis (NSK), and 533 images of healthy corneas. Sample images of each class are shown in Figure 3. This total differs from that reported in the original publication due to a later update to the Figshare repository [12].

B. DATA PROCESSING

Initially, artifact images were removed from the dataset.[1] As illustrated in Figure 2, these samples present excessive brightness or excessive darkness with little to no diagnostically meaningful content. Retaining such images can inject noise into the training process and encourage the model to learn spurious correlations unrelated to corneal microstructures. Next, the images were normalized to have pixel values between zero and one, as shown in Equation 1. After artifact removal, the dataset was reduced to 2,155 grayscale images.

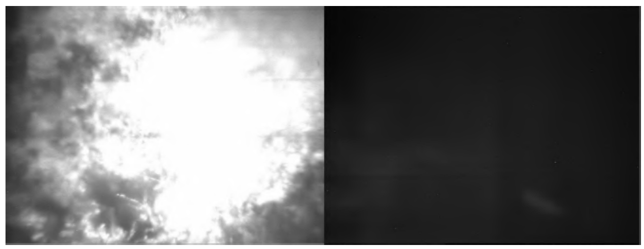

FIGURE 2. Examples of artifact IVCM images removed during dataset curation. The left panel illustrates an overexposed frame with saturated intensity, whereas the right panel shows an underexposed frame with insufficient contrast. Both cases lack reliable morphological information and were excluded to reduce noise and prevent spurious learning.

This exclusion step was performed automatically using a rule-based quality filter to remove frames dominated by illumination artifacts or near-empty content. Specifically, an image was flagged as overexposed when more than 90% of its pixels had intensities in the range [200, 255], which indicates saturation likely caused by direct illumination. Conversely, an image was flagged as underexposed when more than 90% of its pixels fell within [0, 80], indicating insufficient signal and an absence of meaningful corneal structures. Images meeting either criterion were discarded prior to training to reduce label-independent noise and prevent the model from learning spurious correlations from non-informative frames. Importantly, these extreme cases are also not clinically informative for manual inspection, as highly saturated or excessively dark frames are typically non-interpretable and therefore not useful for human assessment of corneal micro-structures. This fully automated procedure ensures consistent and reproducible artifact screening across the dataset.

$$y = \frac{1}{255} \cdot x_{\text{gray}} \tag{1}$$

[1] A total of 863 images were removed, including 500 images belonging to class AK, 263 to class FK, 51 to the healthy cornea class, and 49 to NSK.

In this study, the dataset was initially partitioned into 80% training and 20% testing sets. To ensure rigorous evaluation and prevent data leakage, this split was performed prior to any augmentation or balancing steps.

Due to the fully anonymized nature of the public IVCM-Keratitis dataset [3], which does not specify unique patient identifiers in its documentation or repository, the data split was performed at the image level. To mitigate potential leakage and ensure experimental rigor, the 80/20 partition was strictly executed prior to any data augmentation or balancing steps [3]. This precaution, combined with the automated artifact filtration, ensures that the model learns discriminative morphological patterns rather than memorizing redundant frames or spurious correlations [3].

Following the partition, six data augmentation techniques were applied exclusively to the training set to enhance generalization and feature variability: positive rotation, negative rotation, horizontal flip, vertical flip, color jittering, and light Gaussian blur. These transformations were performed offline and uniformly across all four classes (AK, FK, NSK, and Normal).

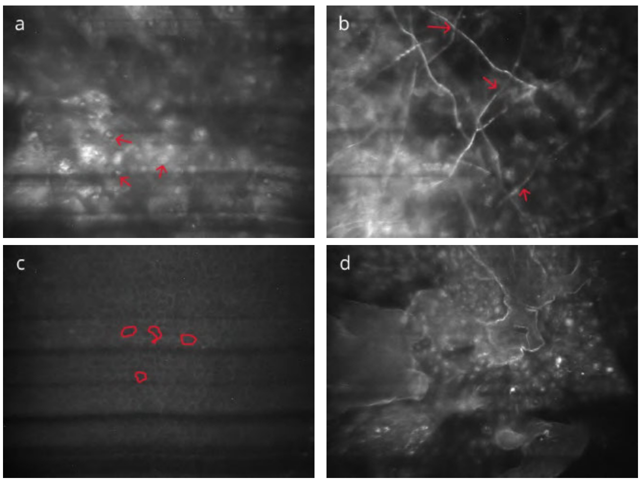

FIGURE 3. Representative IVCM images show the morphological features of the dataset classes. (a) Trophozoite form of Acanthamoeba, with a large nucleolus and contractile vacuoles; (b) Hyphae are thin and long; (c) Healthy and normal corneas; (d) Non-specified keratitis, exhibiting mixed structural features, diffuse opacities, and irregular tissue patterns.

Subsequent to augmentation, the class imbalance affecting the AK, FK, and Normal classes is addressed. Instead of indiscriminate reduction, a targeted random undersampling strategy is applied. Critically, to preserve the intrinsic diversity and feature variance of the classes, all original seed images were retained. Undersampling was applied solely to the synthetic augmented instances to align the total count with the minority class (NSK). This approach ensures that the model remains exposed to the full manifold of the original data distribution while prioritizing the quality and authenticity of features over synthetic volume. Table 1 presents the final class distribution.

TABLE 1. Image distribution by class. The training columns report the number of images before and after class balancing, while the test set is kept unchanged.

Class	Train (Before)	Train (After)	Test
Acanthamoeba Keratitis (AK)	696	1,008	195
Fungal Keratitis (FK)	474	1,008	126
Normal Corneas (Normal)	387	1,008	95
Unspecified Keratitis (NSK)	144	1,008	38
Total	1,701	4,032	454

C. MODEL DEVELOPMENT

A two-stream convolutional neural network pipeline, named CAJU, is proposed to leverage the complementary inductive biases of two distinct backbones by pairing a texture-oriented network with a structure-oriented (or efficiency-oriented) network. Rather than being restricted to a single fixed configuration, CAJU is instantiated and comparatively evaluated using three backbone pairs: DenseNet-161 + EfficientNet-B0, DenseNet-161 + ResNet-101, and VGG16 + MobileNetV3.

The architectural choice for each dual-stream instantiation in CAJU is grounded in the principle of feature specialization. For the DenseNet-161 + ResNet-101 pairing, DenseNet-161 is employed as a texture-oriented stream; its dense connectivity and feature reuse allow it to preserve the high-frequency spatial details necessary for detecting subtle pathogen signatures, such as hyphal fragments or Acanthamoeba cysts. In contrast, ResNet-101 serves as a structure-oriented stream, utilizing its deeper residual blocks to capture broader morphological context and corneal layer deformations.

Similarly, when pairing VGG16 with MobileNet-V3, CAJU explores the synergy between traditional deep-feature extraction and modern hardware-efficient design. This modularity is not a simple architectural trial; it is a deliberate strategy to address the multiscale nature of IVCM findings, where diagnostic evidence ranges from microscopic fungal filaments to macroscopic tissue opacities. The self-attention block acts as the core of this synergy, dynamically recalibrating the contribution of each stream to ensure that the final embedding prioritizes the most reliable morphological cues for each individual case.

As illustrated in Figure 4, each stream processes the same input image. To ensure dimensional compatibility for fusion, a Global Average Pooling (GAP) layer is applied to the final convolutional feature maps of each backbone, transforming them into compact one-dimensional vectors. These vectors are then concatenated to form the joint representation, which is subsequently processed by a SA block.

Inspired by the Transformer architecture [13], this attention mechanism computes dynamic dependencies between the combined features. Unlike standard concatenation, which

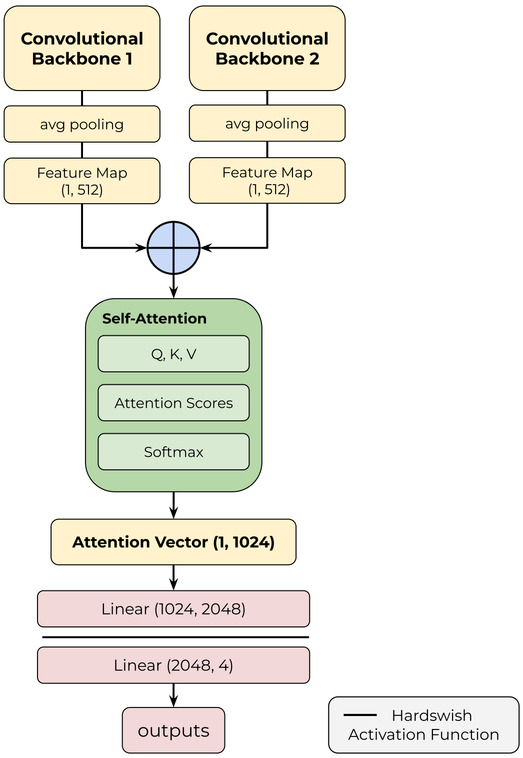

FIGURE 4. Schematic overview of the CAJU dual-stream pipeline. Two convolutional backbones (Backbone 1 and Backbone 2) process the same input image in parallel, enabling the exploration of different complementary backbone pairs (DenseNet-161 + EfficientNet-B0, DenseNet-161 + ResNet-101, and VGG16 + MobileNetV3). Each stream produces a compact feature vector (1 × 512), obtained through pooling to ensure dimensional compatibility. The vectors are concatenated into a joint embedding (1 × 1024) and refined by a self-attention block (Q, K, V). The attended representation is then forwarded through fully connected layers to generate the final class outputs.

treats all features equally, the SA block assigns learnable weights to the fused vector, effectively recalibrating the feature space to emphasize diagnostically relevant patterns while suppressing noise. This enriched representation is then fed into a multilayer perceptron (MLP) for final classification.

Let $\mathbf{x}_1, \mathbf{x}_2 \in \mathbb{R}^d$ denote the d-dimensional embeddings extracted from the two backbone streams. Instead of applying attention to a single vector, we interpret the two embeddings as a length-2 sequence and stack them into a token matrix:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \end{bmatrix} \in \mathbb{R}^{2\times d}. \tag{2}$$

We then compute the query, key, and value projections:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V, \tag{3}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d_a}$ are learnable projection matrices and d_a is the attention dimension. Scaled dot-product self-attention is computed as:

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_a}}\right) \in \mathbb{R}^{2\times 2}, \tag{4}$$

where the softmax is applied row-wise. The attended token representations are:

$$\tilde{\mathbf{X}} = \mathbf{A}\mathbf{V} \in \mathbb{R}^{2\times d_a}. \tag{5}$$

Finally, the attended features are mapped back to the original space and combined with a residual connection.

$$\mathbf{H} = \mathbf{X} + \tilde{\mathbf{X}}\mathbf{W}_O \in \mathbb{R}^{2\times d}, \tag{6}$$

with $\mathbf{W}_O \in \mathbb{R}^{d_a\times d}$. The final fused representation is obtained by concatenating the two attended tokens.

$$\mathbf{h} = [\mathbf{H}_{1,:}; \mathbf{H}_{2,:}] \in \mathbb{R}^{2d}, \tag{7}$$

which is then fed to the MLP classifier to predict keratitis classes.

For model optimization, a 5-fold cross-validation scheme was adopted. Hyperparameter tuning was conducted via grid search over 20 epochs, exploring batch sizes of {8, 16, 32}, learning rates of {0.01, 0.001, 0.0001}, and early stopping patience of {5, 7, 10}. The optimal configuration yielded a batch size of 16, a learning rate of 0.0001, and a patience of 7 epochs. Following hyperparameter selection, the CAJU model and the standalone backbones were trained for up to 100 epochs. To effectively leverage transfer learning and ensure stable convergence, a two-stage training strategy was employed. For the initial 5 epochs, the convolutional layers were frozen to preserve the pre-trained ImageNet representations, allowing only the classifier head to update. Subsequently, the entire network was unfrozen for fine-tuning. Early stopping was utilized throughout the process to halt training once performance had plateaued.

To mitigate the risk of overfitting stemming from the constrained sample size of 2,155 curated images, the CAJU pipeline integrates several regularization and data-centric strategies designed to enhance generalization. First, the adoption of transfer learning with ImageNet-pre-trained backbones provides a foundation of general-purpose visual representations. This approach facilitates faster convergence on domain-specific IK patterns, even when the availability of labeled IVCM data is limited. Second, an data augmentation pipeline was applied exclusively to the training set. This strategy expands the representational manifold and prevents the model from memorizing specific training instances by injecting label-independent variance. Furthermore, the training process was governed by an early stopping mechanism. This ensures the optimization terminates upon reaching the generalization peak, effectively preventing the model from over-adjusting to noise or spurious correlations within the training distribution.

D. EVALUATION

Evaluation was performed exclusively on the held-out test set to measure the model's ability to generalize to truly unseen images. During testing, inference consisted of a single forward pass with all parameters frozen (i.e., no gradient computation or model updates), ensuring a clean separation between training/validation and evaluation and preventing any information leakage.

Given the clinical nature of IVCM diagnosis, performance was quantified using four standard metrics: accuracy, precision, sensitivity (recall), and F1-score. Although accuracy provides an overall summary of correctness, greater emphasis was placed on sensitivity to reflect the model's ability to identify pathological cases and on F1-score to capture the trade-off between precision and sensitivity.

E. ABLATION STUDY

The comparative analysis benchmarked the proposed CAJU dual-stream pipeline against representative standalone backbones employed within the study (e.g., VGG16, MobileNetV3, EfficientNet-b0, DenseNet-161, and ResNet-101), enabling a direct comparison between single-stream baselines and the fused dual-stream representation. Furthermore, to rigorously quantify the contribution of the attention mechanism, an ablation study was conducted by evaluating CAJU in two configurations: (i) with the self-attention (SA) block and (ii) without the SA block, using direct feature concatenation. This controlled isolation enables a direct assessment of the attention module's impact on the discriminative capacity of the final embedding.

F. COMPUTATIONAL COMPLEXITY

We report the number of parameters (in millions), physical model size (in MB), and computational cost in terms of GFLOPs for a single forward pass. All GFLOPs values are computed using an input resolution of 224×224 and a batch size of 1. The model size corresponds to the serialized checkpoint size under the same precision setting used in our experiments. Table 2 summarizes these indicators for CAJU approaches and representative deep learning backbones commonly adopted in medical image analysis.

TABLE 2. Trainable parameters (in Millions, M), model size (in MB), computational cost (GFLOPs), and efficiency ratios (MB/GFLOPs and M/GFLOPs) of CAJU compared to deep learning models.

Model	Params (M)	MB	GFLOPs	MB/G	M/G
MobileNetV3-Large [14]	5.40	21.10	0.22	95.91	24.55
EfficientNet-B0 [15]	5.28	20.50	0.39	52.56	13.53
AlexNet [8]	61.0	233.00	0.72	323.61	84.72
Inception V3 [8]	27.0	104.00	5.70	18.25	4.74
DenseNet161 [3]	28.70	110.40	7.70	14.34	3.73
ResNet101 [7], [9]	44.50	170.50	7.80	21.86	5.71
ConvNeXt-B [16]	88.60	338.10	15.40	21.95	5.75
VGG16 [6]	138.40	528.00	15.50	34.06	8.93
VGG19 [17]	143.7	548.00	19.60	27.96	7.33
VisionTransformer_l_16 [18]	304.0	1161.00	61.55	18.86	4.94
CAJU (DenseNet+ResNet+SA)	30.87	297.49	15.57	7.56	1.98
CAJU (EfficientNet+DenseNet+SA)	37.5	143.13	8.43	16.97	4.44
CAJU (VGG16+MobileNet+SA)	30.9	117.76	15.96	7.37	1.93

Table 2 indicates that the CAJU models offers a favorable balance between capacity, stored model size, and computational cost across the evaluated dual-stream variants. Notably, the fusion stage (self-attention and MLP head) is kept fixed across variants, and only the backbone feature extractors are replaced; therefore, parameter variations across CAJU configurations are expected to be smaller than those observed when comparing against monolithic single-backbone models.

In the VGG16+MobileNetV3 configuration, CAJU operates at a computational cost comparable to widely used CNN baselines (VGG16 and ConvNeXt-B at $\sim$15.5 GFLOPs), while being markedly more compact: it reduces the parameter count from 138.4M (VGG16) to 30.87M ($\approx$77.7% fewer parameters) and decreases the stored model size from 528.00MB to 117.76MB (approximately 77.6% smaller). Among the CAJU variants, DenseNet161+ResNet101+SA provides a higher-storage option, whereas DenseNet161+EfficientNet-B0+SA offers a lower-compute alternative, which may be preferable under tighter latency constraints. Overall, the CAJU variants achieve the most favorable normalized efficiency ratios in Table 2 (lowest MB/G and M/G), supporting attention-based dual-stream fusion as a practical design to preserve representational capacity while keeping deployment requirements controlled. In addition to these complexity measures, practical deployment efficiency is supported by the latency results reported in Table 4, which quantify end-to-end inference time on both GPU and CPU.

III. EXPERIMENTS

The performance metrics for the evaluated architectures are summarized in Table 3. Overall, the CAJU hybrids with self-attention (w/ SA) achieved the strongest results. While CAJU (VGG16 + MobileNetV3, w/ SA) obtained the highest accuracy (97.13% $\pm$ 0.86), CAJU (DenseNet-161 + ResNet-101, w/ SA) delivered the best class-balanced performance, achieving the highest precision (96.22% $\pm$ 0.84), recall (96.01% $\pm$ 1.02), and F1-score (96.03% $\pm$ 1.00). Given the clinical setting and the relevance of precision/recall trade-offs, these results suggest that the DenseNet-161 + ResNet-101 CAJU configuration offers the most reliable overall performance, whereas the VGG16 + MobileNetV3 variant remains attractive when maximizing overall accuracy is a priority.

A detailed generalization analysis is supported by the performance consistency across the folds. As reported in Table 3, the narrow standard deviation ($\pm$1.00% for the CAJU configuration's F1-score) indicates that the model is robust to variations in the data distribution. This stability suggests that the CAJU pipeline effectively captures fundamental morphological patterns, such as the distinctive textures of hyphae and cysts, that are invariant across different data subsets. Furthermore, the high performance maintained on the strictly held-out test set, which was excluded from all optimization phases, provides empirical evidence that the framework generalizes well to unseen clinical cases, mitigating the bias often associated with limited dataset size.

A critical finding emerges from the ablation study across the three CAJU variants. For the VGG16 + MobileNetV3 hybrid, removing self-attention leads to a severe performance collapse (F1-score 76.90% $\pm$ 2.66), substantially below the standalone VGG16 (91.01% $\pm$ 2.62) and MobileNetV3

TABLE 3. Mean (± std) of the evaluated approaches, including standalone backbones, hybrid architectures without self-attention (w/o SA), and hybrid architectures with self-attention (w/ SA).

Model	Accuracy	Precision	Recall	F1
EfficientNet-B0	90.18% ± 3.03	85.09% ± 3.38	84.95% ± 2.97	84.47% ± 3.74
VisionTransformer_h_16	83.72% ± 2.41	83.51% ± 2.56	84.00% ± 2.19	83.71% ± 2.41
DenseNet-161	92.48% ± 2.04	92.94% ± 1.89	92.48% ± 2.04	92.42% ± 2.07
ResNet-101	85.69% ± 0.05	86.85% ± 0.03	85.69% ± 0.05	85.73% ± 0.04
VGG16	94.37% ± 2.27	90.89% ± 2.91	91.79% ± 1.11	91.01% ± 2.62
MobileNetV3	96.04% ± 0.56	93.77% ± 0.79	93.01% ± 1.63	93.30% ± 1.01
EfficientNetB0 + DenseNet-161 (w/o SA)	90.97% ± 1.84	86.34% ± 2.66	86.55% ± 2.59	85.84% ± 2.50
DenseNet-161 + ResNet-101 (w/o SA)	95.42% ± 0.47	92.54% ± 1.14	91.61% ± 1.22	91.93% ± 0.92
VGG16 + MobileNetV3 (w/o SA)	85.07% ± 2.00	77.95% ± 2.56	77.13% ± 2.66	76.90% ± 2.66
EfficientNetB0 + DenseNet-161 (w/ SA)	85.99% ± 1.76	80.28% ± 2.25	79.21% ± 2.52	78.74% ± 1.58
DenseNet-161 + ResNet-101 (w/ SA)	96.01% ± 1.02	96.22% ± 0.84	96.01% ± 1.02	96.03% ± 1.00
VGG16 + MobileNetV3 (w/ SA)	97.13% ± 0.86	94.77% ± 1.28	95.80% ± 1.49	95.24% ± 1.34

(93.30% ± 1.01), indicating that naive fusion can induce strong feature interference when combining heterogeneous representations. In contrast, adding the Self-Attention (SA) block restores and surpasses baseline performance, increasing the F1-score to 95.24% ± 1.34 (+18.34 percentage points), which supports SA as an effective recalibration mechanism for this pairing. Notably, the standalone Vision Transformer (ViT-H/16) achieved an intermediate performance level (F1-score 83.71% ± 2.41), outperforming the degraded EfficientNetB0+DenseNet-161 w/o SA fusion but remaining well below the attention-enabled CAJU configuration, suggesting that attention-driven feature interaction is more beneficial here than relying solely on global token mixing.

The inclusion of the Vision Transformer (ViT-H/16) in our benchmark addresses recent shifts toward purely attention-based architectures [18]. However, as shown in Table 3, the ViT achieved an intermediate F1-score of 83.71% ± 2.41%, falling significantly behind the best CAJU configuration. This performance gap suggests that while global token mixing is powerful, it may struggle to capture the fine-grained, high-frequency morphological cues, such as subtle fungal filaments or Acanthamoeba cysts, that CNN-based streams are natively biased to extract from IVCM images.

Beyond the comparison with purely attention-based models, a similar benefit is observed for DenseNet-161 + ResNet-101, where SA improves F1 from 91.93%±0.92 (w/o SA) to 96.03% ± 1.00 (w/ SA), yielding a +4.10 percentage-point gain and suggesting a more stable synergy between these two high-capacity streams. Conversely, EfficientNet-B0 + DenseNet-161 exhibits a different behavior: the SA version degrades from 85.84% ± 2.50 (w/o SA) to 78.74% ± 1.58 (w/ SA), indicating that attention-based fusion is not universally beneficial and may depend on the complementarity of the paired backbones and the calibration of the fusion module. Overall, the ablation results demonstrate that SA can be a key structural component for effective dual-stream integration, but its impact is architecture-dependent, ranging from substantial gains to performance degradation.

To evaluate the statistical significance of the performance gains, we conducted a paired t-test comparing the F1-scores of the reference CAJU configuration (DenseNet-161 + ResNet-101, with self-attention) against its standalone components (DenseNet-161 and ResNet-101) across the 5-fold cross-validation.

Specifically, given the sample size ($n = 5$ folds), the 95% confidence interval (CI) was computed using Student's t-distribution with $df = n - 1 = 4$ degrees of freedom. The CI was calculated as:

$$\mathrm{CI} = \bar{x} \pm t_{\alpha/2,\, df}\, \frac{s}{\sqrt{n}}, \tag{8}$$

where $\bar{x}$ is the mean F1-score (96.03%), s is the standard deviation (1.00%), and $t_{\alpha/2,\, df}$ is the critical value (≈ 2.776) for a 95% confidence level.

This assessment yielded $p < 0.05$ in all comparisons, confirming that the integration of self-attention provides a statistically significant improvement over single-stream baselines. The resulting 95% CI for the reference model's F1-score was [94.79%, 97.27%], further indicating the framework's predictive stability.

Synthesizing these statistical insights with the results in Table 3, the subsequent analyzes focus on the best-performing configuration, CAJU (DenseNet-161 + ResNet-101, w/ SA). This model achieved the strongest class-balanced performance, with the highest precision, recall, and F1-score, indicating a consistently reliable trade-off between false-positive and false-negative rates across cross-validation folds. To visualize the training dynamics and further validate the effectiveness of the regularization framework, Figure 5 presents the learning curves (categorical cross-entropy loss and validation F1-score) for this configuration across all five folds. The stable convergence observed across all partitions confirms that the integration of transfer learning, extensive data augmentation, and early stopping effectively mitigated overfitting risks despite the dataset size. Given the clinical relevance of F1-score behavior and the need for robust generalization, CAJU (DenseNet-161 + ResNet-101, w/ SA) is adopted as the reference model for the following experiments and interpretability analyzes.

A. HARDWARE LATENCY

To ensure a fair and reproducible comparison, all latency experiments were conducted on a single machine equipped with an NVIDIA GeForce RTX 3050 GPU (6 GB VRAM) and an Intel Core i5-13420H CPU with 32 GB RAM. The evaluated models correspond to CAJU pipelines (dual-backbone architectures) incorporating the self-attention (SA) block. Inference latency was measured with batch size = 1, and we report the mean ± standard deviation (ms/image) over 200 forward passes after an initial warm-up period, separately for GPU and CPU execution. The results are summarized in Table 4.

Table 4 shows that the CAJU VGG16+MobileNetV3+SA pipeline is the most efficient, achieving the lowest latency

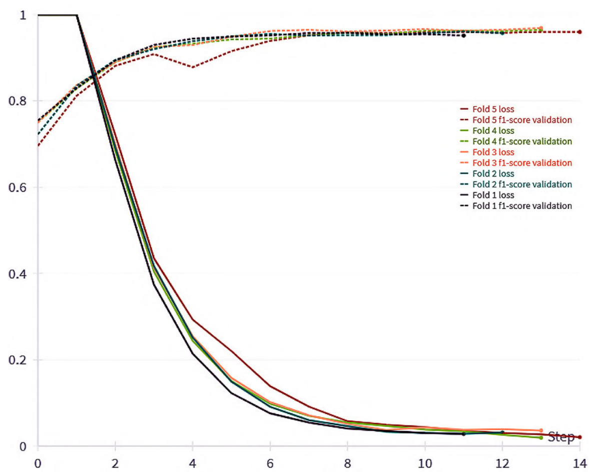

FIGURE 5. **Learning curves for the CAJU (DenseNet-161 + ResNet-101, w/ SA) pipeline across 5-fold cross-validation. The training loss (solid lines) and validation F1-score (dashed lines) demonstrate stable convergence and high consistency between folds, supporting the model's robustness and effective mitigation of overfitting.**

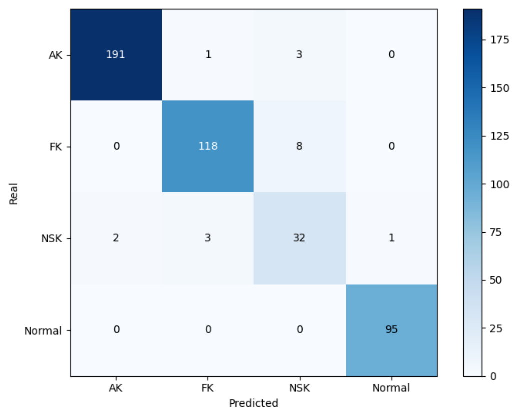

FIGURE 6. **Confusion matrix of the CAJU pipeline on the held-out test set, instantiated with DenseNet-161 + ResNet-101 with Self-Attention (w/ SA) and trained using the optimal hyperparameters on the full training split. Rows indicate ground-truth labels and columns indicate predicted labels for the keratitis classes (AK, FK, NSK, and Normal).**

TABLE 4. **Inference latency for SA-based dual-backbone models (mean ± std).**

Model	GPU (ms/image)	CPU (ms/image)
DenseNet+EfficientNet+SA	23.07 ± 5.19	136.57 ± 6.55
VGG16+MobileNetV3+SA	$\mathbf{7.34 \pm 0.70}$	$\mathbf{101.80 \pm 3.16}$
DenseNet+ResNet+SA	23.58 ± 5.90	187.66 ± 2.59

on both GPU (7.34 ± 0.70 ms/image) and CPU (101.80 ± 3.16 ms/image). The DenseNet+EfficientNet+SA and DenseNet+ResNet+SA configurations are slower on GPU (around 23 ms/image), and DenseNet+ResNet+SA presents the highest CPU latency (187.66 ± 2.59 ms/image). Overall, the results indicate a clear efficiency advantage for the VGG16+MobileNetV3 backbone pairing under the same SA fusion strategy.

B. EXPLICABILITY

To assess diagnostic reliability at the patient-facing level, Figure 6 reports the confusion matrix on the held-out test set for the CAJU pipeline instantiated with its best-performing pairing, DenseNet-161 + ResNet-101 with Self-Attention (w/ SA). The model maintained strong recognition of the most distinctive categories, with high sensitivity for AK and Normal. Most errors are concentrated in the clinically challenging inflammatory classes, particularly involving NSK.

Importantly, the remaining errors concentrate on a clinically plausible morphological ambiguity between FK and NSK. The dominant confusion pattern corresponds to FK cases predicted as NSK, with the reciprocal error also present. This cluster suggests that, in challenging cases, fragmented hyphal structures and low-contrast fungal signatures may resemble non-specific inflammatory infiltrates typical of NSK, thereby hindering discrimination based on texture cues alone. Consistent with this observation, NSK remains the most difficult class, reinforcing that the primary obstacle for automated diagnosis lies in separating pathogen-specific patterns from generic inflammation, rather than in recognizing healthy tissue or the more structurally distinctive AK patterns.

To enhance interpretability and assess clinical plausibility, we incorporated complementary explainability methods, combining Grad-CAM [11] and SHAP [19] to visualize the evidence driving the CAJU pipeline's decisions. Since the selected CAJU configuration relies on a dual-stream backbone (**DenseNet-161 + ResNet-101, w/ SA**), independent Grad-CAM heatmaps were generated for each stream to inspect their spatial contributions, while SHAP was used to provide class-wise attributions on the input space, enabling a more fine-grained analysis of the decision rationale.

In the AK classification example (Figure 7), the explanations are consistent with a clinically plausible decision. The Grad-CAM maps obtained from the two CAJU streams (DenseNet-161 and ResNet-101) show a high degree of spatial agreement, concentrating their strongest responses on near bright, granular cluster in the central region of the IVCM image, possible a pattern compatible with cyst-like structures.

This convergence suggests that the dual-stream representation does not rely on diffuse background artifacts, but rather on localized morphological cues. Complementarily, the SHAP analysis indicates that the most influential attributions are confined to this same discriminative area, while the attribution maps for the non-target classes remain comparatively weak, supporting the notion that the prediction is driven by localized evidence instead of global intensity variations.

The FK case (Figure 8) further illustrates the interpretability and clinical plausibility of the CAJU pipeline when instantiated with DenseNet-161 + ResNet-101 (w/ SA). In this correctly classified example, the Grad-CAM maps highlight high-contrast, filament-like patterns and adjacent hyper-reflective regions that are consistent with fungal morphology

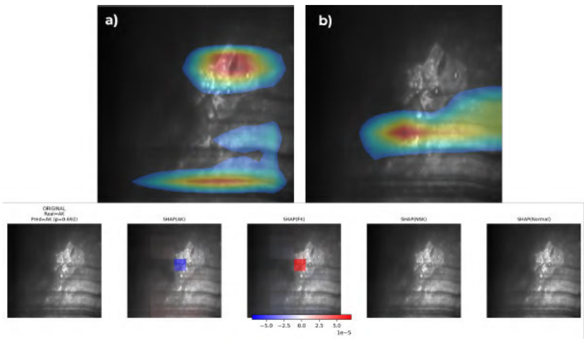

FIGURE 7. Explainability outputs for a correctly classified AK case. Top row: Grad-CAM heatmaps for the CAJU pipeline instantiated with DenseNet-161 (a) and ResNet-101 (b), highlighting the image regions most influential to each stream. Bottom row: SHAP class-wise attribution maps for AK, FK, NSK, and Normal, illustrating the localized evidence supporting (or countering) each class decision.

in IVCM. Although both streams focus on diagnostically relevant structures, their responses are complementary: one backbone produces more localized peaks over salient foci, whereas the other yields a broader activation over the lesion extent, capturing contextual evidence around the elongated pattern.

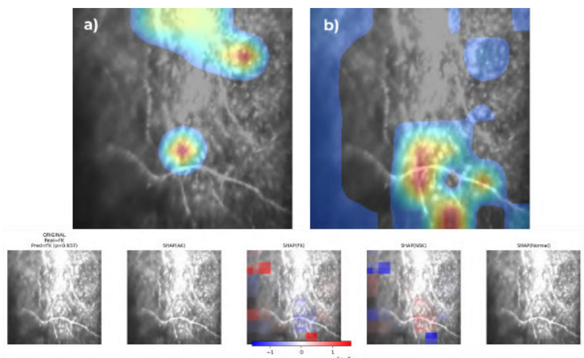

FIGURE 8. FK prediction case explained by Grad-CAM and SHAP for the CAJU pipeline DenseNet-161 + ResNet-101, w/ SA. Top row: Grad-CAM heatmaps for the CAJU pipeline instantiated with DenseNet-161 (a) and ResNet-101 (b) highlighting filament-like and hyper-reflective regions associated with fungal patterns in IVCM. Bottom: class-wise SHAP attributions, where the FK map concentrates positive evidence on the same discriminative structures, while alternative classes (AK, NSK, Normal) receive weaker or opposing contributions.

This interpretation is reinforced by the SHAP analysis, which provides class-wise attributions at the input level. The SHAP(FK) map concentrates positive contributions (red) on regions overlapping the salient structures emphasized by Grad-CAM while showing negative contributions (blue) in less informative areas. In contrast, the attribution map for competing classes (SHAP(NSK)) exhibits weak and predominantly negative evidence around the same regions, suggesting that the decision is driven by class-discriminative morphological cues rather than diffuse background intensity. Overall, the agreement between Grad-CAM localization and SHAP class-wise attributions supports the notion that the CAJU pipeline bases FK predictions on clinically meaningful features and resolves the classification using a combination of focal and contextual information.

We include a correctly classified NSK example to illustrate how the CAJU pipeline behaves in an intrinsically ambiguous scenario (Figure 9). In this case, the specific Grad-CAM maps of the DenseNet-161 and ResNet-101 flows focus on distinct regions of hyper-reflective infiltrate, producing broad but coherent activations over the main pathological area rather than a single point clue. This behavior is expected for NSK, which is typically characterized by non-pathognomonic inflammatory patterns and limited pathogen-specific structures.

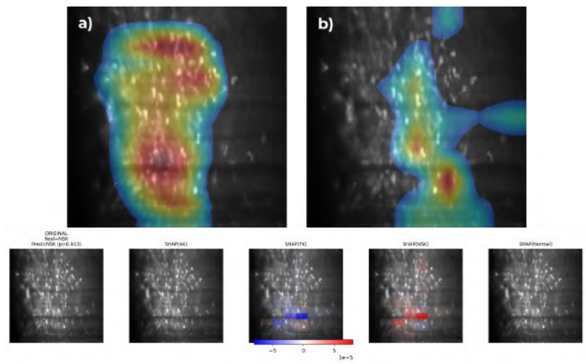

FIGURE 9. Correct NSK prediction explained by Grad-CAM and SHAP for the CAJU pipeline, DenseNet-161 + ResNet-101, w/ SA. Top row: stream-specific Grad-CAM maps highlighting the main hyper-reflective infiltrate region. Bottom: class-wise SHAP attributions, where SHAP(NSK) concentrates positive evidence on the abnormal texture band, while SHAP (FK) shows predominantly negative contributions over overlapping regions, supporting discrimination from fungal-like patterns.

the SHAP maps reveal competing evidence: while SHAP(NSK) assigns positive contributions to the central abnormal texture band, SHAP(FK) also presents localized positive attributions over partially overlapping regions. This pattern suggests that the image contains morphological cues compatible with both nonspecific inflammation and fungal-like textures, which is consistent with the FK and NSK confusion observed in the confusion matrix. Nevertheless, the aggregate evidence remains stronger for NSK, indicating that the model considered FK as a plausible alternative hypothesis but ultimately favored the nonspecific diagnosis.

To better characterize the main error observed in the confusion matrix, Figure 10 illustrates a representative misclassification in which an FK sample was predicted as NSK. The Grad-CAM visualizations of the two streams focus on broad infiltration regions but do not consistently isolate a distinct filament-like pattern typically associated with FK. Instead, the activated areas overlap with regions that can also be explained by inflammatory textures, cysts, or even other pathogens, which are characteristic of NSK and constitute a clinically plausible source of ambiguity.

Beyond the overlap observed in Fig. 6, FK/NSK confusion has clinically asymmetric implications. FK predicted as NSK is a false negative for fungal keratitis, which can delay

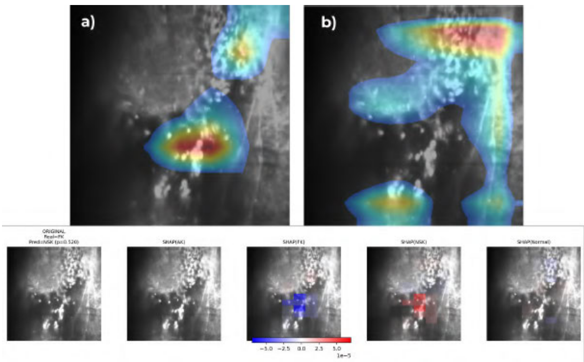

FIGURE 10. Misclassification example explained by Grad-CAM and SHAP for the CAJU pipeline DenseNet-161 + ResNet-101, w/ SA. Top: stream-specific Grad-CAM maps highlighting diffuse attention over hyper-reflective infiltrate regions. Bottom: class-wise SHAP attributions showing positive evidence for NSK over the lesion area and negative contributions for FK, consistent with FK and NSK morphological overlap.

appropriate antifungal management and allow progression if treated as nonspecific keratitis. NSK predicted as FK is a false positive suspicion of FK, which may trigger unnecessary antifungal exposure and additional confirmatory testing. This pattern is plausible in IVCM because fragmented or low-contrast fungal elements and diffuse hyper-reflective infiltrates can resemble nonspecific inflammatory phenotypes grouped as NSK, especially in low-signal or visually ambiguous frames.

The SHAP attributions reinforce this interpretation by revealing competing evidence at the input level. While SHAP(NSK) assigns positive contributions to patches covering the main infiltrate region, SHAP(FK) shows weak and partially negative attributions over overlapping areas, suggesting that the features available in this image are more aligned with the nonspecific inflammatory phenotype learned for NSK than with pathogen-specific fungal signatures. Importantly, the moderate confidence of the prediction ($p = 0.520$) indicates uncertainty rather than overconfident failure, supporting that morphological overlap remains the dominant challenge for automated diagnosis in difficult cases.

In order to mitigate the clinical risk of FK being predicted as NSK, it is recommended that the model be employed as a diagnostic aid. The application of uncertainty-aware triage entails the flagging of low-confidence predictions or minimal top-two margins between FK and NSK. It is imperative that these cases be prioritized for specialist review and/or laboratory confirmation. This approach has the potential to facilitate the distribution of effort more effectively by expediting cases where diagnoses are more certain and focusing attention on cases where presentations are ambiguous.

IV. DISCUSSION

Despite technological advances in laboratory diagnostics, effective management of infectious keratitis (IK) remains hindered by the limitations of conventional methods. Traditional microbiological tests exhibit low sensitivity and significant time delays [3], while molecular techniques like PCR, although offering higher sensitivity (75-100% for FK and 67-75% for AK [2]), are restricted by high costs and the need for specialized infrastructure. In this context, IVCM emerges as a powerful, non-invasive alternative. However, its clinical utility is currently bottlenecked by the requirement for highly specialized expertise to interpret subtle morphological features, making manual analysis prone to inter-observer variability and diagnostic errors [3].

The proposed CAJU pipeline aims to reduce this gap by automating IVCM interpretation through attention-guided dual-stream feature fusion. Rather than defining a single fixed model, CAJU provides a modular framework in which different backbone pairings can be instantiated according to performance and deployment constraints. The best-performing instantiation, DenseNet-161 + ResNet-101 with Self-Attention (w/ SA), achieved the strongest class-balanced performance (highest precision, recall, and F1-score), indicating robust discrimination across keratitis subtypes. This finding supports the core hypothesis that combining complementary representations and explicitly recalibrating them via attention yields a more informative embedding than relying on a single stream alone, particularly in a setting where clinically relevant cues may appear at different spatial scales and contrast levels.

While [10] reported slightly higher metrics, direct comparison is limited by differences in datasets, acquisition protocols, and class distributions. Nevertheless, the general principle of learning complementary representations and integrating them via feature-fusion mechanisms informed the design of the CAJU pipeline. Importantly, our results indicate that fusion quality is instance-dependent: although attention improves performance for some backbone pairings, it may not universally benefit all combinations, reinforcing that backbone complementarity and the calibration of the fusion module are critical factors.

Beyond predictive performance, CAJU was designed with clinical transparency in mind. To this end, we incorporated explanation methods based on **Grad-CAM** [11] and SHAP [19] to provide spatial and class-wise attribution cues that support clinical plausibility checks. In correctly classified cases, the explanations highlight localized regions consistent with pathogen-related patterns. In contrast, for the challenging NSK cases, SHAP often reveals competing evidence between inflammatory textures, aligning with the dominant confusion pattern observed in the confusion matrix. These properties support the use of CAJU as a scalable decision-support tool for IVCM, potentially expanding access to expert-level interpretation in non-specialized centers.

Deep neural networks have catalyzed a paradigm shift in medical image analysis, delivering strong performance across diverse modalities such as X-ray, retinal fundus, MRI, and CT [20]. In the specific domain of IK, convolutional neural networks have been extensively explored to automate etiological classification, motivated by the need to reduce inter-observer variability and accelerate time-to-diagnosis in routine workflows.

Early approaches to automated IVCM analysis primarily adopted single-stream CNNs, where architectures such as DenseNet [3], ResNet [7], [9], [21], and AlexNet [6] established strong baselines for IK classification. Building on these foundations, more recent studies have investigated hybrid strategies to capture complementary cues that may not be fully represented by a single backbone. In particular, [10] combined distinct CNN families to leverage architectural diversity, supporting the broader observation that integrating heterogeneous representations can improve diagnostic performance when the fused features are properly calibrated.

In parallel with performance gains, there is increasing emphasis on XAI for medical imaging applications. As noted in prior work [10], visual explanations are essential to mitigate the abstracted perception of deep learning and enable plausibility checks aligned with clinical reasoning. Accordingly, CAJU incorporates explanation mechanisms (Grad-CAM and SHAP) to provide both spatial localization and class-wise attribution, facilitating the interpretation of correct predictions and the systematic inspection of challenging error modes (NSK ambiguity).

To the best of our knowledge, this study is among the first to integrate a self-attention fusion module within a dual-stream pipeline tailored for IK classification from IVCM images. In contrast to simple feature fusion, the proposed attention mechanism dynamically recalibrates the contributions of each stream, enabling the fused representation to emphasize diagnostically informative patterns while attenuating less relevant responses from background stromal texture. Importantly, our ablation results show that the impact of attention is architecture-dependent: it provides substantial gains for some pairings (VGG16 + MobileNetV3 and DenseNet-161 + ResNet-101) but may not improve all combinations, highlighting the role of backbone complementarity and fusion calibration hyperparameters.

When instantiated with the best-performing configuration, CAJU (DenseNet-161 + ResNet-101, w/ SA), the pipeline achieved the highest class-balanced performance (precision = 96.22%, recall = 96.01%, and F1-score = 96.03%). These results support the notion that attention-guided fusion acts as a feature selector, strengthening discriminative cues associated with pathogen-specific morphology while reducing interference from non-informative regions, thereby improving both diagnostic reliability and interpretability when accompanied by explanation methods (Grad-CAM and SHAP).

While these studies provide the foundational landscape for automated IK diagnosis, direct performance benchmarking is constrained by inter-study variability, particularly regarding dataset composition and experimental pipelines. A detailed summary of these related works is provided in Table 5.

Despite the robust performance achieved across multiple cross-validation folds, we explicitly acknowledge the absence of external validation on independent datasets as a current limitation of this study.This constraint arises from the fact that the IVCM-Keratitis dataset remains the only publicly available open-access resource for the etiological classification of infectious keratitis. While our results demonstrate strong internal consistency and architectural robustness, the model's sensitivity to domain shifts, such as variations in confocal hardware, acquisition protocols, and diverse clinical populations, remains to be fully characterized.

TABLE 5. Related works on automatic keratitis classification using IVCM images. For our pipeline, Dataset Size denotes the number of real IVCM images retained after artifact filtering (2,155 images; 1,701 train + 454 test). The training split was further augmented and balanced, resulting in 4,032 training samples.

Model	Dataset Size	Results	Additional Techniques
DenseNet161 [3]	4,001	Accuracy 96.93% Precision 92.52% Sensitivity 94.77% F1-score 93.93%	Grad-CAM Transfer learning
ResNet101 [9]	68,970	Accuracy 94.6%	Transfer learning
SACNN [10]	7,278	Accuracy 97.73% Precision 98.68% Specificity 98.54% Sensitivity 97.02% F1-score 97.84%	Feature fusion
ResNet101 [7]	2,088	Accuracy 96.26% Specificity 98.34% Sensitivity 91.86% AUC 98.75%	Transfer learning
AlexNet [6]	1,213	Accuracy 99.95% Sensitivity 99.90% Specificity 100.0%	Contrast enhancement Histogram matching
ResNet101 [8]	3,453	Accuracy 96.5% Sensitivity 93.6% Specificity 98.2% AUC 98.3%	Grad-CAM Guided Grad-CAM
CAJU (DenseNet-161 + ResNet-101, w/ SA)	2,155	Accuracy 96.01% Precision 96.22% Sensitivity 96.01% F1-score 96.03%	Grad-CAM Transfer learning

To mitigate this, future work will focus on establishing multi-center collaborations to perform prospective evaluations in real-world clinical settings, ensuring that the CAJU pipeline maintains its predictive power across different imaging environments. This is particularly relevant as the present experiments were conducted on the public IVCM-Keratitis dataset, which was collected from a large clinical pool of IVCM exams from hundreds of patients acquired at the Confoscan unit of the Central Eye Bank of Iran. [3].

V. CONCLUSION

Across the evaluated configurations, CAJU pipelines with self-attention achieved the strongest overall results. In particular, the best-performing instantiation, DenseNet-161 + ResNet-101 (w/ SA), delivered the highest class-balanced performance, while VGG16 + MobileNetV3 (w/ SA) achieved the highest accuracy. These findings support the notion that attention-guided fusion improves discriminative reliability for heterogeneous morphological patterns.

To support clinical transparency, the proposed pipeline was complemented with explanation strategies based on **Grad-CAM** and **SHAP**, enabling spatial localization and class-wise attribution analyzes for both correct predictions and clinically relevant error modes (notably NSK ambiguity). Although not intended to replace expert ophthalmologists, CAJU shows potential as a decision-support tool to reduce inter-observer variability and assist non-specialized centers in triaging and etiological assessment.

Future work will focus on validating generalization under real-world domain shifts, expanding the etiological spectrum (bacterial and viral keratitis), and incorporating uncertainty-aware calibration to better manage ambiguous cases and support safe clinical deployment.

CODE AVAILABILITY

The code for the CAJU model developed in this study is publicly available at the following GitHub repository: https://github.com/IsaacRamos1/CAJU_model

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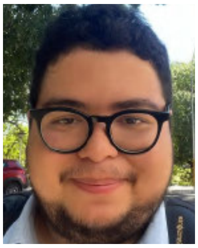

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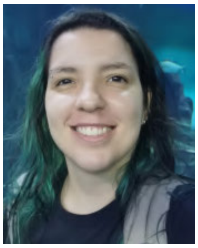

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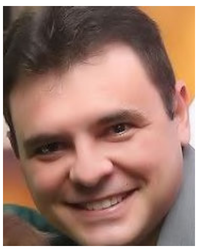

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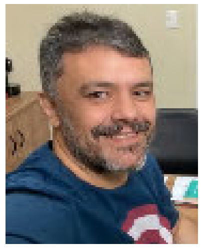

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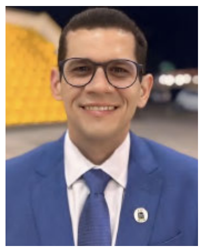

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